

Various Landforms

Mainly there are three types of landforms—Mountains, Plateaus, Plains.

Mountains

The height of mountains are over 600 m and have conical peaks. On the basis of origin there are four types of mountains : Block Mountains, Residual Mountains, Accumulated Mountains and Fold Mountains.

Block Mountains

1. The middle part of such mountains is lower and the parts on both the sides are higher. The middle lower portion is called as Rift valley. The longest rift valley is the valley of the Jordan river.
2. Black Forest (Germany), Vindhyachal and Satpura (India), Salt Range (Pakistan) are some examples of block mountains.

Residual Mountains

Such mountains are formed as a result of weathering. Examples— Aravalli, Nilgiri, Parasnath, Hills of Rajmahal (India), Siera (Spain).

Accumulated Mountains

These are formed due to accumulation of sand, soil, rocks, lava etc. on the Earth's Crust., e.g. Sand Dunes.

Fold Mountains

These are formed because of the folds in the rocks due to internal motions of the earth. These are wavelike mountains which have numerous peaks and lows, e.g. Himalayas, Ural, Alps, Rockies, Andes etc.

Plateaus

Plateaus are extensive upland areas characterised by flat and rough top surface and steep walls which rise above the neighbouring ground surface at least for 300 m.

Generally the height of plateau ranges from 300 to 500 feet.

Inter mountainous Plateaus : Plateaus formed between mountain, Example-Tibetan Plateau.

Mountain step Plateaus : The flat region between a plain and the base of a mountain.

Continental Plateaus : These are formed when the Lacolith inside the Earth comes to the surface due to weathering, e.g. the Southern Plateau

Bank Plateaus : These are the plateaus on the banks of the oceans.

Domelike Plateaus : These are formed due to the movement of man and animals on the surface, e.g. Ramgarh Plateau.

Plains

Plains can be defined as flat areas with low height (below 500 ft.)

Weathered Plains : The plains formed due to weathering by rivers, glaciers, winds etc.

Loess Plains : These are formed by the soil and sands brought by winds.

Karst Plains : Plains formed due to the weathering of limestone.

Erosional Plains : Plains near the river banks formed by river erosion.

Glacial Plains : Marshy plains formed due to the deposition of ice.

Desert Plains : These are formed as a result of the flow of rivers.

Deposition Plains : Large plains are formed due to the silt brought by the rivers. Such plains are plains of Ganga, Sutlej, Mississippi, Hwang Ho.

Forests

They are of the following types :

(a) Tropical Evergreen Rain Forests : Such forests are found in the equatorial and the tropical regions with more than 200 cms annual rainfall. The leaves of trees in such forests are very wide. Ex— Red wood, palm etc.

(b) Tropical Semi Deciduous Forests : Such forests receive rainfall less than 150 cms. Saagwan, saal, bamboo etc. are found in such forests.

(c) Temperate mixed Forests : Such forests are a mixture of trees and shrubs. Corks, Oak etc. are the major trees of these forests.

(d) Coniferous Forests or Taiga : These are evergreen forests. The trees, in these forests, have straight trunk, conical shape with relatively short branches and small needlelike leaves. Example—Pine, Fir etc.

(e) Tundra Forests : Such forests are covered with snow. Only Mosses, a few sludges and Lichens grow here in the summers. This type of vegetation is chiefly confined to the northern hemisphere (e.g. in Eurasia, North America and Greenland Coast).

(f) Mountainous Forests : Vegetation varies according to altitude.

Pastures (or Grasslands)

They can be divided into two types :

(i) Tropical Pastures and

(ii) Temperate Pastures

(i) Tropical Pastures : They have different names in different countries. Savanna in Africa, Campos in Brazil, Llanos in Venezuela and Columbia.

(ii) Temperate Pastures : They are known by the following names—Prairies in USA and Canada, Pampas in Argentina, Veld in South Africa, Rangelands or Downs in Australia and New Zealand, Steppes in Eurasia (Ukraine, Russia).

Land forms created by the river system

V-shaped valley

1. A river flows with a greater velocity in the mountainous region and big, pointed fragments of rock also flow with a great speed along with the water.
2. The river bed is scoured and downcutting starts, ultimately giving rise to a deep valley with steep sides. This valley is called a v-shaped valley.
3. These valleys are found in mountainous regions.
4. A deep and narrow valley with steep sides is called a gorge.
5. The gorge of the river Ulhas in Thane district in Maharashtra and the gorge of the river Narmada at Bhedaghat near Jabalpur in Madhya Pradesh are well known.

6. There are many gorges in the Himalayas.

Waterfall

1. If there are both hard (resistant) and soft (less resistant) rocks in the course of the river, the less resistant rock is eroded faster.
2. The resistant rock does not erode so easily. That is why, the river falls with a great speed from a cliff-like part of hard rock. This is called a waterfall.
3. The Niagara Falls on the Niagara river is in North America.

Potholes

1. In areas where the river bed consists of hard rock, the stones carried along with the river water due to the whirling impact of water.
2. That is why holes of various shapes are formed in the rocky river bed. Such holes are called potholes.
3. Many patholes are observed in the river bed of the Kukadi, Krishna, Godavari etc. in Maharashtra.

Meanders and ox-bow lakes

1. Meanders are formed by lateral erosion. As the erosion increases over a period of time, the meanders in the river again starts flowing in a straight line.
2. The loop previously formed then separates from the main course of the river. Water accumulates in this separated part.
3. As this loop resembles on ox-bow it is called ox-bow lake. It formed due to impounding of water in the abandoned meander loop.

Fan-shaped plains

1. In the region near the source of a river the tributaries joining the main river deposit materials carried by them on the banks of the main river.
2. This deposition creates fan-like plains. They are called fan-shaped plains or alluvial fans.

Flood plains

1. When, during the floods, the river-water overflows its banks and spreads in the surrounding areas, the silt carried by the water gets deposited in those areas. This creates flat plains on both the banks of the river. Plains created by this depositional work done during floods are called flood plains.
2. The Gangetic Plain is a flood plain.

Natural levees

1. When a river is over flooded, its water crosses its banks. At that time, the speed of the water is reduced, and the pebbles and stones carried by the river get deposited near the banks.
2. On account of frequent floods, the area where these sediments are deposited near the bank of the river rises higher than the flood plain. This high wall is called a natural levee or natural embankment.
3. Such levees are found on the banks of the Mississippi, the Huang-ho etc., Southern bank of river Ganga.

Delta

1. Delta was coined by Herodotus (the 'Father of History') after the Greek letter delta (Δ) because of the deltoid shape at the mouth of the Nile.
2. A delta is a land form that is formed at the mouth of a river where that river flows into an ocean, sea, estuary, lake, reservoir, flat arid area or another river.
3. Deltas are formed from the deposition of the sediment carried by the river as the flow leaves the mouth of the river. Over long periods of time, this deposition builds the characteristic geographic pattern of a river delta.

Delta-region

1. A river meets a sea or a lake. The silt carried by the river is deposited on the bed near its mouth.
2. The area near the mouth of the river gets gradually filled up by this deposition and gets raised causing an obstruction for the river to flow in a single channel. It, therefore, splits into two branches and meets the sea.

3. Over a period of time, there is deposition also at the mouth of the sebranches. In this manner, the main course of the river gets split into a network of small channels. These sub-channels are called distributaries.
4. A triangular region of innumerable such distributaries is formed near the mouth of the river. This region is called the delta region.
5. There are delta regions near the vent (opening) of the rivers Godavari, Ganga, Nile, Mississippi etc. Deltas are very fertile.
6. The largest delta of the world is 'Sunderbans Delta' (350 km.).

Glacier

1. A mass of ice sliding down the slope from a snow-clad region is called a glacier. On an average a glacier moves 1 to 15 metres a day.
2. While a glacier is moving, the friction of the ice at the bottom slows down the movement of the bottom layers.
3. There are two main types of glaciers : (i) Continental Glacier and (ii) Alpine Glacier.

Continental Glacier

1. An extensive sheet of ice spreading across a vast region sometimes begins to move due to the pressure of the ice.
2. This moving sheet of ice is called a continental glacier.
3. Such glaciers are seen in Antarctica and Greenland.

Alpine or mountain glacier

1. There are snow-field in the mountainous regions of the Himalayas, the Alps, the Andes, the Rocky mountains etc.
2. The ice accumulating in these areas starts sliding down the slopes.
3. This mass of ice sliding down from the mountains is called a mountain glacier or an alpine glacier.

Iceberg

1. Blocks of ice break off from the continental glaciers and float away into the sea.
2. A block of ice floating in the sea is called an iceberg. These icebergs are huge in size.
3. The density of ice being slightly less than that of water, a very little portion of an iceberg is seen above the water and the rest of it is submerged under water.

Land forms of glaciation

1. Various land forms are created on account of the transportation, erosion and depositional work of a glacier. Let us consider the major land forms thus created.

Cirque

1. When the snow from the mountain peaks slides, it gets deposited in a hollow, if there is one on any side of the peak.
2. The accumulated snow starts sliding down the slope. This causes friction at the floor and at the sides of the hollow, thus enlarging it further. This is called a cirque.
3. The back wall of a cirque is like a high cliff and the floor is concave and huge in size. The total shape resembles an armchair.
4. When a glacier melts completely, water accumulates in the cirque and forms a lake which is known as tarn.

Fiord

1. Where the lower end of the trough is drowned by the sea it forms a deep steep-side inlet called 'Fiord' as on the Norwegian and South Chilean Coasts.

U-shaped valley

1. When a glacier is flowing through a valley in a mountainous region, the sides of the valley get eroded. Ice causes friction on the sides of the valley.
2. As the erosion of the sides is greater than that of the floor, a valley is formed with vertical sides and a wide floor. This valley is called a U-shaped valley.

Hanging valley

1. In the mountainous region, many tributaries join the main glacier.

2. The quantity of ice in a tributary is comparatively smaller. Hence, it causes less friction.

3. The valley of a tributary is at a higher level than a valley of the main glacier, the valley of the tributary appears to be hanging. That is why, such a valley is called a hanging valley.

Moraine

1. The material transported and deposited by a glacier is known as moraine.

2. Moraines are made up of pieces of rocks that are shattered by frost action and are brought down the valley.

Moraines are of the following types

(1) lateral moraine, (2) medial moraine (3) terminal moraine and (4) ground moraines.

1. After a glacier has melted, different land forms of deposition are seen.

2. The oval-shaped hills of lesser height are called drumlins.

3. Zig-zag hills, with many steep slopes, made up of long stretches of sand and gravel are called eskers.

Land forms created by the action of wind

Mushroom rock

1. The wind blowing in desert regions erodes the rock near the ground surface to a great extent. At the same time, the upper part of the rock gets eroded to a lesser extent.

2. As this is a continuous process, the foot of the rock becomes narrow.

3. The top portion of the rock then looks like an umbrella. This land form is called a mushroom rock.

Sand dunes

1. Sand gets transported from one place to another along with the wind.

2. At a spot where the wind meets an obstruction or where the speed of the wind reduces, dunes are formed out of the sand which gets deposited.

3. The side of the dune facing the wind has a gentle slope and the opposite side has a steep slope.
4. Because of the slow speed of the wind, the sand on the gentle slope gets carried to the top and comes down the steep slope on the other side. Sand dunes gradually move forward in this manner.

Barkhan

1. The fine sand particles carried by the wind get deposited when the speed of the wind is reduced forming crescent shaped dunes. Such hills are called barkhans.

Loess

1. Loess is a soil finer than sand.
2. Loess is a silt transported by the wind from the desert regions and deposited much further way.
3. Loess transported from the desert regions of Central Asia has been deposited in layers in China. The plain they form is known as the Loess plain.

Groundwater

1. Some water from the rainfall received on the earth's surface seeps through the ground.
2. This water trickles down until it reaches an impervious rock.
3. Water accumulated under the ground surface in this manner, is called ground water.
4. Some rocks on the earth's surface are porous and some have cracks or joints. Water seeps in through these pores or joints.
5. Groundwater gushes out in the form of springs.

Land forms created by the actions of groundwater

Sink holes

1. Water on the ground surface seeps through limestone. Some portion of the limestone dissolves in that water. If this process takes place continuously, it makes holes in these rocks.

2. As this process continues over a number of years, these holes get enlarged. These holes are called sink holes.

Caves

1. In limestone region, water goes very deep through sink holes.
2. If there is a layer of impervious and hard rock underneath, water flows horizontally on the impervious rock instead of going deeper.
3. Hence, soft rocks get eroded and a cave is formed.

Stalactites and stalagmites

1. Inside the cave created by groundwater under the ground surface in a limestone region, water is always seeping through the roof. This water contains calcium carbonate.
2. As the seeping water evaporates, some of the calcium carbonate, it contains, is deposited on the cave's roof. This deposition continues to grow very slowly. Hence a column is seen growing from the roof towards the floor. It is called a stalactite.
3. The water dripping on the floor of the cave also evaporates leaving behind calcium carbonate which accumulates over a period of time.
4. A column then starts growing from the floor to the roof. This column which grows upwards is called a stalagmite.
5. Stalactites and stalagmites are observed in the Parner Taluka of Ahmadnagar district, in Bastar District in Chhattisgarh and also in the Karst region of former Yugoslavia now Serbia and Montenegro.

Land forms created by the actions of sea waves

Sea Cliff

1. The base of the rocks on the coast get eroded because of the impact of the ocean waves and notches develop in these rocks.
2. The crest of the rock overhangs the notch. These notches in the rocks gradually extend landwards over a period of time. Then the crest falls and a steep cliff, which has receded away from the sea is formed.

Sea cave

1. Rocks on the coast have many cracks. They become wider and wider with the impact of the waves, creating small caves. They are called sea caves.
2. Such sea cliffs and sea caves are observed at Shrivandhan, Ratnagiri, Malvan, Vengurle etc.

Beach

1. The fine sand and other material that flows along with the waves get deposited in a direction parallel to the sea coast.
2. This deposition of sand is called a beach.
3. There are extensive beaches in the coastal regions of the states of Maharashtra, Goa, Kerala, Tamil Nadu, Odisha and West Bengal in India and in other countries like Bangladesh and Canada.

Sand bar

1. A deposition of sand which results in a long, narrow embankment in the sea near the coast is called a sand bar.

Lagoon

1. A shallow lake is formed between the sand and the sea coast. It is called a lagoon. Such a lake is called Kayal in Kerala.